



1 - PURPOSE

The rationale for a wet granulation process for the production of high dose niacin lipid matrix sustained release tablets was demonstrated in part I of this study. In part II the effects of process parameters on granule growth of niacin: Compritol® 888 blends, subsequent tablet properties and drug release rates are evaluated.

2 - INTRODUCTION

Part I of the study describes a wet granulation process with niacin in a Compritol® 888 ATO matrix using a mortar and pestle. In part II the process parameters are evaluated in a fluid bed granulator (FBG) with focus on granule growth, physical properties and drug release. The identification of critical parameters is crucial to implement PAT (Process Analytical Technology) according to the principles of Quality by design [1].

3 - EXPERIMENTAL METHODS

600g of Compritol® 888 ATO and niacin ratio 1:1 were blended with 3% (w/w) PVP for 5 min at 90 rpm, fluidized for 5 min and granulated in a Glatt GPCG1.1 at varying processing conditions (Table 1). Granules were dried until the product reached the outcome temperature, and moisture content measured. PVP was added in different concentrations or as 3% binder solution. Particle size distribution and granule properties were analyzed prior to tableting (Korsch EK0) with 0.5% magnesium stearate as lubricant. Dissolution studies were conducted in 0.1N HCl (USP II, 900 mL, 100 rpm, 37°C). Drug dose was 500mg.

Table 1: Investigated process parameters (niacin:Compritol® 1:1)

Parameter	3% PVP						10% PVP	20% PVP
	FBG 1	FBG 2	FBG 3 (A/B)	FBG 4	FBG 5	FBG 6	FBG 7	FBG 8
Inlet temperature, °C	40	40	40	65	65	65	40	40
Product temperature, °C	25	24	24	41	35	36	30	32
Drying temperature, °C	45	45	45	65	65	65	65	65
Spray rate, g/min	8	16	24	24	24	24	24	24
Atomization pressure, bar	1	1	1	1	1	2	2	2
Granulation liquid (water), %	17	17	17	34	17	34	34	34

4 - RESULTS AND DISCUSSION

Table 1 describes all the granulation conditions (FBG1 to 8). All 8 processing conditions resulted in particle growth compared with physical mixture (Figure 1). Increasing atomization pressure in combination with higher water concentration (FBG3 vs. FBG6) and higher PVP concentration (FBG6 vs. FBG7 vs. FBG8) affected granule size. The critical parameters for the niacin:Compritol® formulation and the effect on granule and tablet properties are listed in Table 2.

Table 2: Critical parameters and effects on granule/tablet properties.

Critical parameters	FBG 3A	FBG 3B	FBG 6	FBG 7	FBG 8
Atomization pressure, bar	1	1	2	2	2
PVP CONCENTRATION, %	3	3 (solution)	3	10	20
Granule properties					
Mean particle size, µm	99.2 (±53.4)	112.7 (±57.3)	134.5 (±69.5)	313.6 (±160.3)	448.3 (±230.9)
Carr's Index, %	20	15	16	12	12
Bulk density, g/cm³	0.4	0.46	0.43	0.46	0.46
Hausner ratio	1.26	1.17	1.18	1.14	1.14
Tablet characteristics					
Weight variation, %	0.35	0.16	0.3	0.22	0.19
Tensile strength, MPa	1.0	0.95	1.1	0.94	0.72
Friability, %	0.57	0.58	0.66	0.74	0.77

Granule growth increased with increasing PVP concentration due to improved powder wetting (Figure 2). The optimum flowability was obtained with 10% of PVP and remained unchanged with 20% of PVP, despite further granule growth. Tablets made with larger granules had the lowest weight variation, but also the lowest tablet hardness (due to increased bulk density and reduced surface area available for inter-particulate bonding). The addition of PVP in the binder solution resulted in larger granules with better homogeneous size distribution compared to the addition of dry PVP to the powder blend. This is again likely to improved wetting (Figure 2). Tablets made with larger granules (FBG7 & FBG8) showed slower release rates compared to tablets made with smaller particles (FBG3A, FBG 3B, Figure 3). The process parameters (FBG 3 vs FBG 6) had no significant influence using 3% of dry PVP (Figure 4).

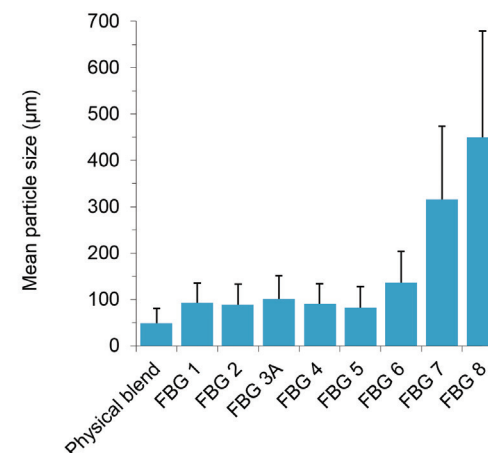


Figure 1: Effect of process parameters on mean granules size.

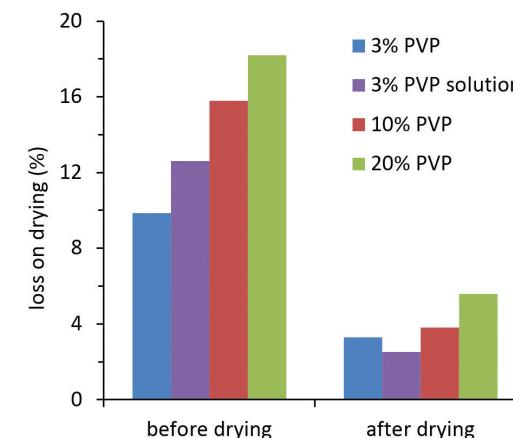


Figure 2: Effect of PVP concentration on LOD of granules.

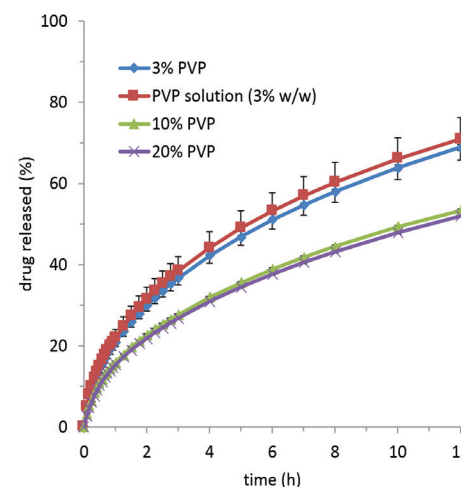


Figure 3: Effect of PVP concentration on drug release.

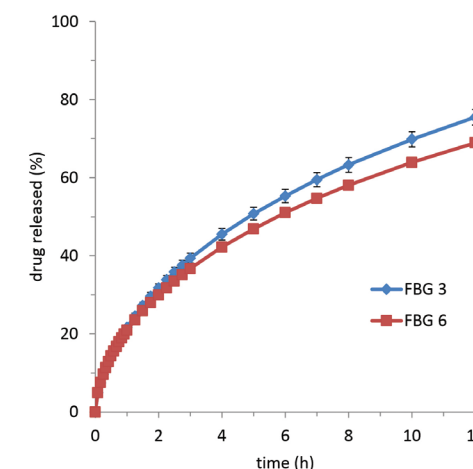


Figure 4: Effect of process parameters on drug release.

5 - CONCLUSION

Granule growth was affected by higher atomization pressures and PVP amounts. Tablet properties and drug release were markedly affected by binder concentration and only slightly by process conditions. Process conditions in FBG3B, FBG6 and 7 produced optimum granules (good flow and compressibility). These findings can be useful during formulation development and construction of the design space following the principles of QbD.

6 – REFERENCES